Properties of Logarithms
Answer Key
Algebra 2

Quick Answers

1. 5
2. $\log(x) + 3\log(2)$
3. 2
4. 2
5. $\frac{\log(x)}{\log(2)} - 5$
6. 5
7. $\log(200)$
8. 6
9. 21
10. 1.681, 1.681
11. $\log(2)$
12. 3

Curriculum

Level: Algebra 2
HSF-BF.B.5 – *problems 1, 2, 3, 4, 5, 7, 8, 10, 11*
HSF-LE.A.4 – *problems 6, 9, 12*
Difficulty 1–4 of 5 across 13 tagged checks.

1. Evaluate using the product property. $\log_2 8 + \log_2 4$

Solution: The product property runs backwards here: a sum of logs with the same base is the log of the product.

$$\log_2 8 + \log_2 4 = \log_2(8 \cdot 4) = \log_2 32$$

Since $2^5 = 32$, the value is five.

$\Rightarrow$ $\boxed{5}$

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2. Expand into separate terms, writing the constant term as a multiple of $\ln 2$. $\ln(8x)$

Solution: Product property first, then the power property on the constant:

$$\ln(8x) = \ln 8 + \ln x = \ln(2^3) + \ln x$$

$\Rightarrow$ $\boxed{3\ln 2 + \ln x}$

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3. Write as a single logarithm, then evaluate. $2\log_3 6 - \log_3 4$

Solution: Power property on the coefficient, then the quotient property:

$$2\log_3 6 - \log_3 4 = \log_3(6^2) - \log_3 4 = \log_3\left(\frac{36}{4}\right)$$

The quotient is nine, and $3^2 = 9$.

$\Rightarrow$ $\boxed{\log_3 9 = 2}$

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4. Evaluate using the product property. $\log_6 4 + \log_6 9$

Solution: Combine first, then recognise the power of the base:

$$\log_6 4 + \log_6 9 = \log_6(4 \cdot 9) = \log_6 36$$

Since $6^2 = 36$, the value is two.

$\Rightarrow$ $\boxed{2}$

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5. Expand using the quotient property. $\log_2\left(\dfrac{x}{32}\right)$

Solution: The quotient property splits a fraction into a difference:

$$\log_2\left(\frac{x}{32}\right) = \log_2 x - \log_2 32$$

and $\log_2 32 = 5$ because $2^5 = 32$, so the second term is a plain number.

$\Rightarrow$ $\boxed{\log_2 x - 5}$

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6. Solve for x. $\log_3 x + \log_3 4 = \log_3 20$

Solution: Condense the left side with the product property, then use that the logarithm is one-to-one:

$$\log_3(4x) = \log_3 20 \implies 4x = 20$$

Dividing by four gives five, and $x = 5$ makes both logarithms' arguments positive, so it is not extraneous.

$\Rightarrow$ $\boxed{x = 5}$

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7. Write as a single logarithm. $3\ln 2 + 2\ln 5$

Solution: Power property on each coefficient, then the product property:

$$3\ln 2 + 2\ln 5 = \ln(2^3) + \ln(5^2) = \ln(8 \cdot 25)$$

$\Rightarrow$ $\boxed{\ln 200}$

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8. **(a)** Expand $\log_2\left(\dfrac{x^5}{16}\right)$. **(b)** Evaluate it at $x = 4$.

Solution (a): Quotient property first, then the power property on the numerator:

$$\log_2\left(\frac{x^5}{16}\right) = \log_2(x^5) - \log_2 16 = 5\log_2 x - 4$$

$\Rightarrow$ $\boxed{5\log_2 x - 4}$

Solution (b): At $x = 4$, $\log_2 4 = 2$, so the expanded form gives $5(2) - 4$.
$\Rightarrow$ $\boxed{6}$

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9. Solve for x. $\log x - \log 3 = \log 7$
Solution: Condense the left side with the quotient property:

$$\log\left(\frac{x}{3}\right) = \log 7 \implies \frac{x}{3} = 7$$

Multiplying by three gives twenty-one, which is positive, so it is a genuine solution.
$\Rightarrow$ $\boxed{x = 21}$

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10. Using only $\log 2 \approx 0.3010$, $\log 3 \approx 0.4771$ and the properties, approximate $\log 48$.
Solution: Factor 48 into powers of the two numbers whose logs are known: $48 = 2^4 \cdot 3$. Now the product and power properties do the rest:

$$\log 48 = \log(2^4) + \log 3 = 4 \log 2 + \log 3 \approx 4(0.3010) + 0.4771$$

That sum is $1.2040 + 0.4771 = 1.6811$, which rounds to three decimal places as follows.
$\Rightarrow$ $\boxed{1.681}$

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11. **(a)** Write $3 \ln x - \ln y - 2 \ln z$ as a single logarithm. **(b)** Evaluate at $x = 2$, $y = 4$, $z = 1$.
Solution (a): Power property on every coefficient first, then one product and one quotient step. Both subtracted logs go into the denominator:

$$3 \ln x - \ln y - 2 \ln z = \ln(x^3) - \ln y - \ln(z^2)$$

$\Rightarrow$ $\boxed{\ln\left(\frac{x^3}{yz^2}\right)}$

Solution (b): Substituting gives $\dfrac{2^3}{4 \cdot 1^2} = \dfrac{8}{4}$, and the logarithm of that quotient is the exact answer below.
$\Rightarrow$ $\boxed{\ln 2}$

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12. Solve for x. $3 \log_2 x = \log_2 27$
Solution: Power property turns the coefficient into an exponent, then match arguments:

$$3 \log_2 x = \log_2(x^3) = \log_2 27 \implies x^3 = 27$$

The only real cube root of twenty-seven is three, and it is positive, so the domain condition holds.
$\Rightarrow$ $\boxed{x = 3}$